

🛒 E-Commerce Analytics | End-to-End Data Engineering Project

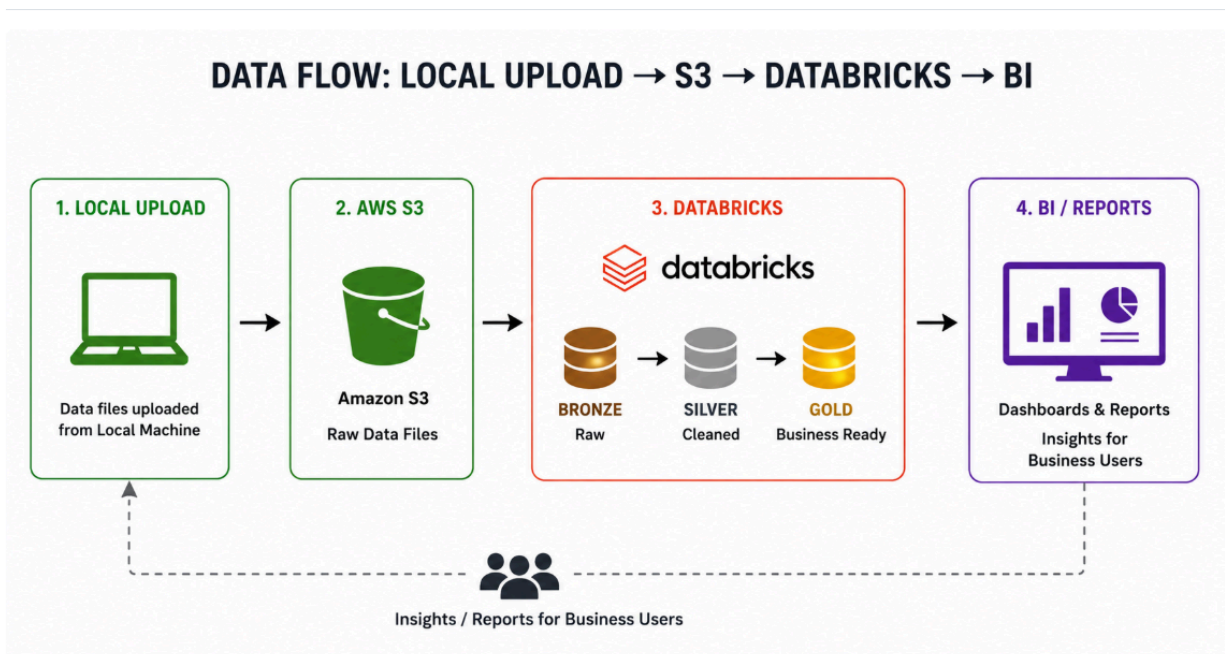
📌 Overview

This project demonstrates an end-to-end **Data Engineering pipeline** built on **Databricks Lakehouse Platform** using the **Medallion Architecture (Bronze → Silver → Gold)**.

The pipeline ingests raw e-commerce transactional data from **AWS S3**, processes it using **Apache Spark**, stores it in **Delta Lake**, and creates business-ready **Dimension** and **Fact** tables for analytics.

The project replaces traditional Python-based ETL scripts with a scalable distributed processing architecture and provides AI-powered analytics through **Databricks Genie** and interactive BI dashboards.

🏗️ Architecture



🚀 Tech Stack

Technology	Purpose
Databricks Free Edition	Lakehouse Platform
Apache Spark	Distributed Data Processing
AWS S3	Data Lake Storage
Delta Lake	ACID Transactions & Time Travel
Unity Catalog	Data Governance
Spark SQL	Analytics
Databricks SQL	Reporting
AI/BI Dashboard	Visualization
Genie AI	Natural Language Query
Databricks Jobs	Pipeline Automation

📁 Project Structure

...

project_ecommerce/

```
├── 1_setup/
│   └── setup_catalog
├── 2_medallion_processing_dim/
│   ├── 2_dim_silver
│   └── 3_dim_gold
├── 3_medallion_processing_fact/
│   └── 3_fact_gold
├── dashboards/
└── README.md
```

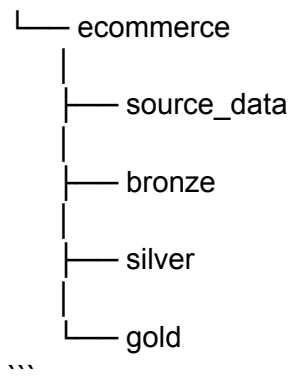
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📁 Catalog Structure

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Catalog

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Catalog



 Serverless Starter Warehouse **Serverless** 2XS

 Catalog ▾



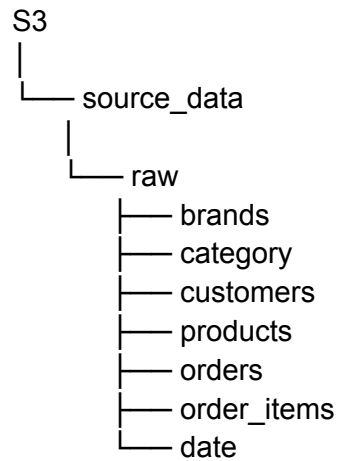
Type to search



- ▼  ecommerce
 - ▼  bronze
 - ▼ Tables (6)
 -  brz_brands
 -  brz_calendar
 -  brz_category
 -  brz_customers
 -  brz_order_items
 -  brz_products
 - >  default
 - ▼  gold
 - ▼ Tables (5)
 -  fact_transactions_denorm
 -  gld_dim_customers
 -  gld_dim_date
 -  gld_dim_products
 -  gld_fact_order_items
 - >  information_schema
 - ▼  silver
 - ▼ Tables (6)
 -  slv_brands
 -  slv_calendar
 -  slv_category
 -  slv_customers
 -  slv_order_items
 -  slv_products
 - ▼  source_data
 - ▼ Volumes (1)
 -  raw

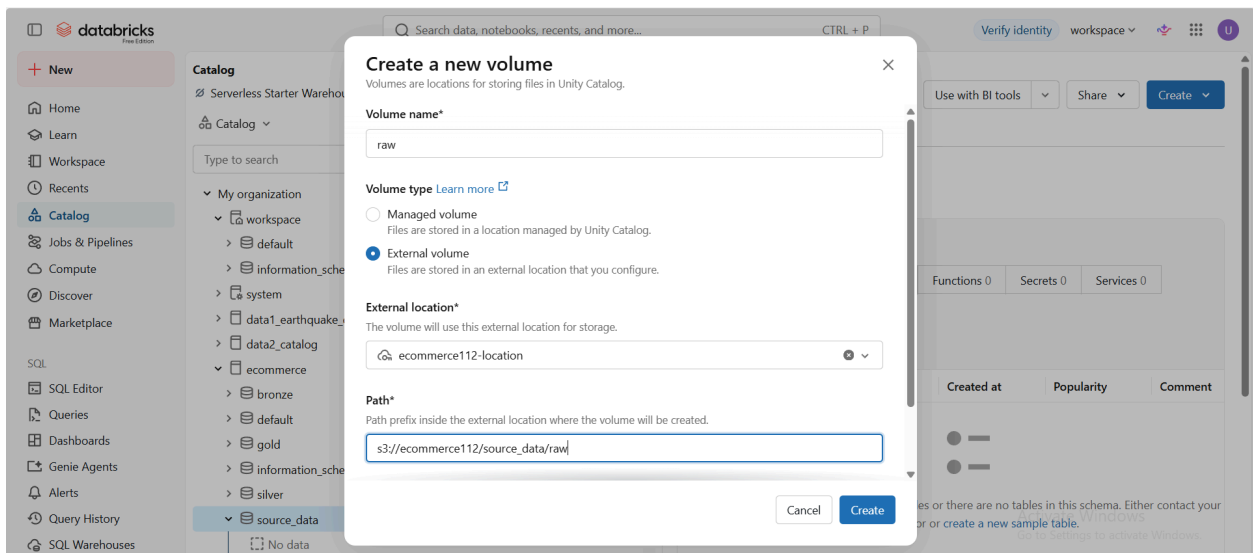
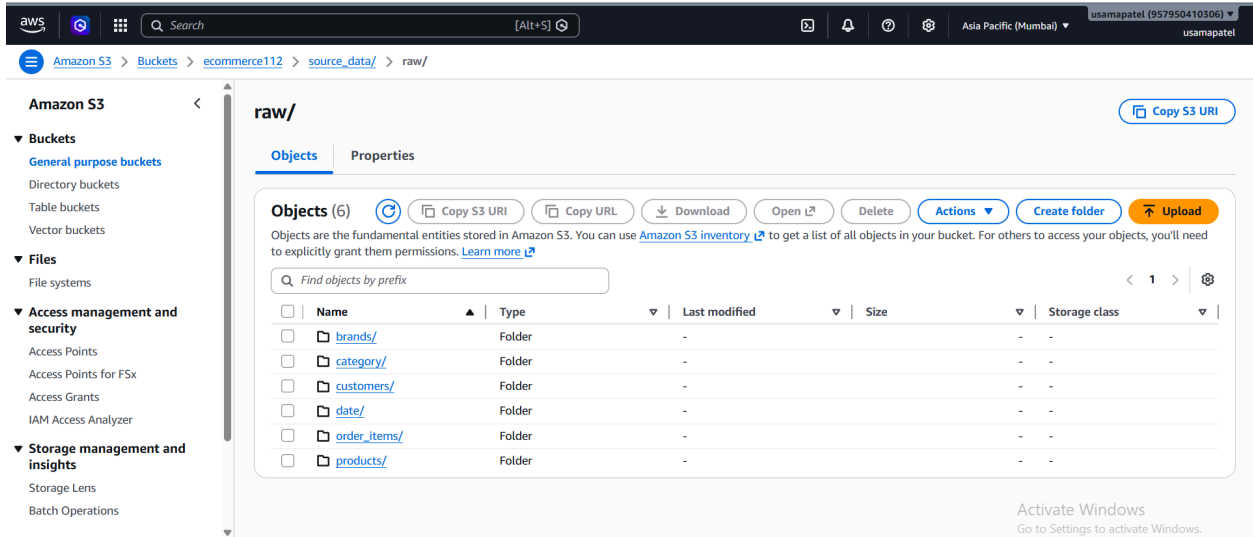
🍷 Data Source

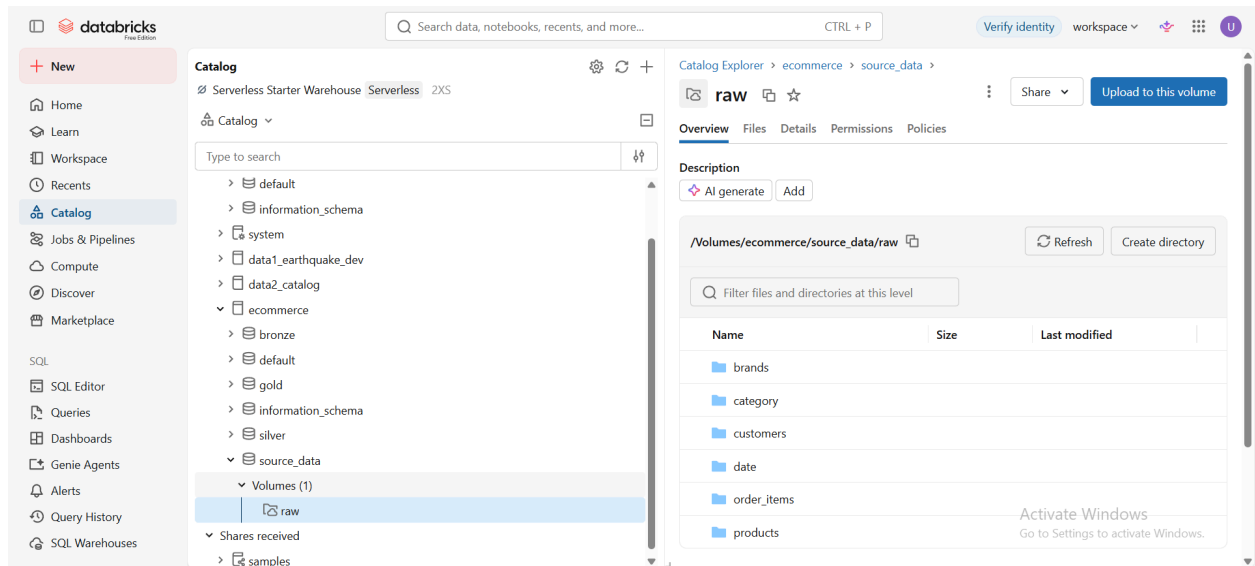
Raw CSV files are stored inside AWS S3.



The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links for Identity and Access Management (IAM), Access Management, and Access reports. The main content area displays the details for a specific IAM role. The 'Summary' section includes the creation date (July 11, 2026, 12:49 UTC+05:30), the ARN (arn:aws:iam::957950410306:role/databricks-uc-7474651308033798-ecommerce112-location), and the maximum session duration (1 hour). The 'Permissions' section shows that the role has one attached policy: 'databricks-ext-loc-access-policy-74746513...'. The 'Filter by Type' dropdown is set to 'All types'. The 'Attached entities' column shows 0 entities.

Policy name	Type	Attached entities
databricks-ext-loc-access-policy-74746513...	Customer inline	0





🏆 Bronze Layer

The Bronze layer stores raw source data without modifying the business data.

Operations

- Raw CSV ingestion
- Schema enforcement
- Delta table creation
- Metadata tracking
- Source file tracking

Metadata Added

- `\_source\_file`
- `\_ingested\_at`

Bronze Tables

- brz_brands
- brz_category
- brz_customers
- brz_products
- brz_orders
- brz_order_items
- brz_date

🏆 Silver Layer

The Silver layer cleans and standardizes the raw data.

Transformations

- Remove duplicate records
- Handle NULL values
- Data type conversion
- Remove special characters
- Standardize text
- Generate business-friendly columns
- Schema validation

Silver Tables

- slv_brands
- slv_category
- slv_customers
- slv_products
- slv_orders
- slv_order_items
- slv_date

🏆 Gold Layer

The Gold layer contains analytics-ready data optimized for reporting.

Dimension Tables

Product Dimension

Contains

- Product ID
- SKU
- Brand

- Category
- Color
- Size
- Material
- Weight
- Rating

Customer Dimension

Contains

- Customer Information
- Demographics
- Country
- Region

Date Dimension

Contains

- Year
- Quarter
- Month
- Week
- Day
- Weekend Flag

Fact Table

fact_transactions_denorm

Denormalized transaction table containing ****34 business columns****.

Transaction Information

- transaction_id
- transaction_date
- transaction_ts

- customer_id
- product_id
- seq_no
- channel

Financial Metrics

- quantity
- gross_amount
- discount_percent
- discount_amount
- tax_amount
- net_amount
- net_amount_inr

Product Attributes

- sku
- category_code
- category_name
- brand_code
- brand_name
- color
- size
- rating_count

Time Dimensions

- year
- quarter
- month_name
- week
- day_name
- hour_of_day
- is_weekend

Promotion Details

- coupon_code
- coupon_flag

Medallion Pipeline

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AWS S3



External Volume



Bronze



Silver



Gold



Dashboards



Business Users

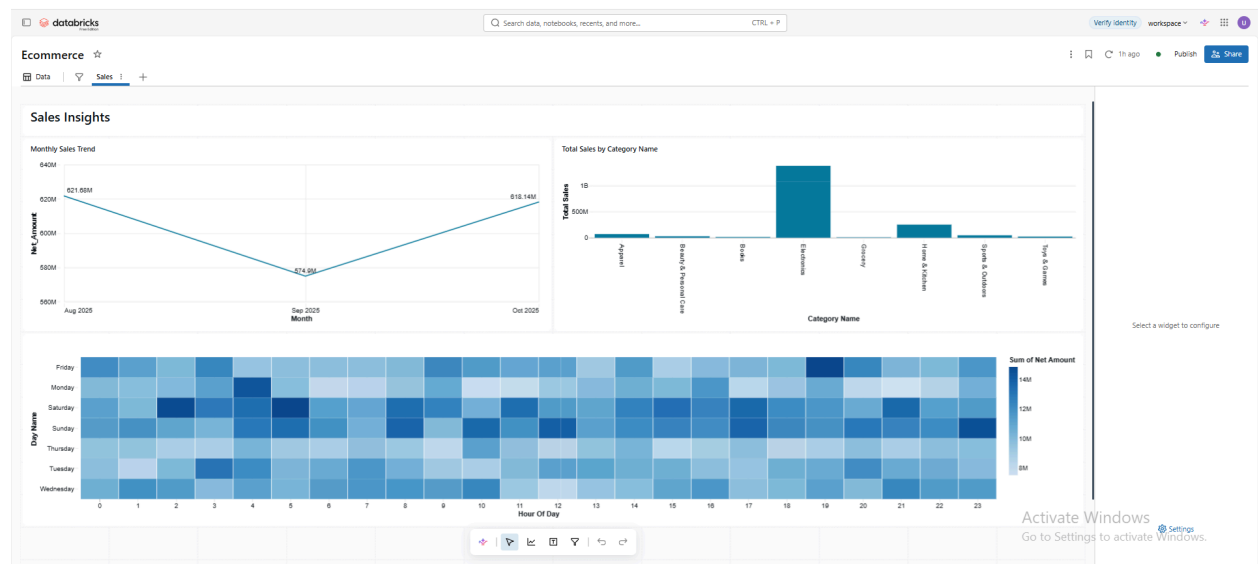
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Dashboard KPIs

The dashboard provides insights into

- Total Revenue

- Total Sales
- Total Orders
- Monthly Sales Trend
- Revenue by Category
- Revenue by Brand
- Channel Performance
- Coupon Performance
- Customer Purchase Analysis
- Product Performance



📊 Business Metrics

Total Sales

₹1.81 Billion

Top Performing Category

Electronics

₹1.38 Billion

Other Categories

- Home & Kitchen
- Apparel
- Sports & Outdoors
- Beauty & Personal Care
- Toys & Games
- Grocery
- Books

💰 Currency Support

Supported currencies

- INR
- USD

Automatic currency conversion to INR is implemented in the Gold layer.

📱 Sales Channels

- Website
- Mobile

🤖 Genie AI

The Gold layer is connected with ****Databricks Genie**** allowing business users to ask natural language questions.

Example

- Show monthly sales.
- Which category generated highest revenue?
- Top 10 selling products.
- Best customers by revenue.
- Sales trend for Electronics.

Data Governance

Implemented using Unity Catalog.

Features

- Catalog Management
- Schemas
- External Volumes
- Managed Tables
- Data Lineage
- Fine-grained Access Control

Orchestration

Pipeline execution is automated using ****Databricks Jobs****.

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S3 Upload

↓

Bronze

↓

Silver

↓

Gold

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Dashboard Refresh

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🌟 Key Features

- End-to-End Data Engineering Pipeline
- Medallion Architecture
- Apache Spark Processing
- AWS S3 Integration
- Delta Lake
- Unity Catalog
- Delta Tables
- Dimension & Fact Modeling
- Currency Conversion
- AI-powered Analytics
- Interactive Dashboards
- Databricks Jobs Automation

🎯 Business Use Cases

- Sales Analytics
- Customer Behavior Analysis
- Revenue Analysis
- Product Performance
- Brand Performance
- Category Analysis
- Coupon Effectiveness
- Order Analytics
- Executive Reporting

🌐 Future Enhancements

- Delta Live Tables
- Auto Loader
- Streaming Pipeline
- Customer Cohort Analysis
- Customer Lifetime Value (CLV)
- Product Recommendation Engine
- Inventory Analytics
- ML Demand Forecasting
- CI/CD with GitHub Actions
- Terraform Deployment

 Author

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****Built using Databricks Lakehouse Platform, Apache Spark, AWS S3, Delta Lake and Unity Catalog.****